

Ecotoxicity

Ecotoxicity refers to the adverse impact of chemicals on the natural environment, including plants, animals, and microorganisms. It involves effects on population growth, reproduction, and overall health of organisms, as well as the consequences on ecosystems and the benefits they provide. The ecotoxicity of a chemical can be influenced by its physical and chemical characteristics, its ability to persist in the environment and its capacity to bioaccumulate.

Table 1. QSAR Flex Ecotoxicity models

Model Name	Version	Model Description	Training Set Size	Descriptors*	Output Units
Fathead Minnow 96h LC50	2.0	Acute toxicity to <i>Pimephales promelas</i> (96 hrs. of exposure)	920	LogP, WS, MW, FRAGS	mg/L
Daphnia 48h LC50	2.0	Acute toxicity to <i>Daphnia magna</i> (48 hrs. of exposure)	2124	LogP, WS, BCF, MW, FRAGS	mg/L
Tetrahymena 48h GC50	2.0	Acute toxicity to <i>Tetrahymena pyriformis</i> (48 hrs. of exposure)	1898	LogP, WS, MP, BCF, MW, FRAGS	mg/L
Algae 72h EC50	2.0	Acute toxicity to various algae (72 hrs. of exposure)	1377	LogP, WS, MP, MW, FRAGS	mg/L
Bio Concentration Factor	2.0	Ratio of concentration of contaminant in organism to surrounding water	563	LogP, WS, VP, FRAGS	Log BCF
Ready Biodegradability	2.0	Aerobic biodegradation potential of a chemical substance within 28 days, as per OECD Test 301.	1443	MW, FRAGS	Binary
Soil Adsorption	2.0	The organic carbon-sorption coefficient (Koc)	651	LogP, WS, BP, VP, FRAGS	Log L/Kg

*LogP – Octanol-water partition coefficient; WS – Water solubility; BCF – Bio concentration factor; MP – Melting point; BP – Boiling point; VP – Vapor pressure; MW – Molecular weight; FRAGS – Molecular fragments.

Model Details

Acute toxicity to Fathead minnow: The Fathead minnow, also referred to as *Pimephales promelas*, is a commonly used model organism in laboratory and field studies to study aquatic toxicology and ecotoxicology. The current model is based on data collected from various publications and the toxicity is expressed in terms of LC50, which is the molar concentration required to cause the death of 50% of the fish within 96 hours. The assay typically involves exposing the fish to different concentrations of a test substance for 96 hours. After this period, the number of dead fish is counted and LC50 is determined.

QSAR Flex Model Information

This model has been trained on a dataset of 920 compounds, with toxicity levels represented as -Log(mol/L). Ordinary least square regression was used as the modeling methodology.

Acute toxicity to Daphnia: *Daphnia magna*, commonly known as the water flea, is widely used in ecotoxicology studies to evaluate the impact of chemicals and pollutants in freshwater ecosystems. It is a useful model organism due to its short life cycle, rapid reproduction, ease of handling, and sensitivity to a wide range of pollutants. The current model, based on data from various publications, expresses toxicity in terms of LC50, the molar concentration that causes 50% mortality within 48 hours. The model was trained on 2124 compounds with toxicity levels represented as -Log(mol/L) and the data was modeled using ordinary least square regression method.

Acute toxicity to Tetrahymena: *Tetrahymena pyriformis* is a commonly found freshwater protozoan, often used as a model organism in toxicology to evaluate the toxic effects of chemicals, drugs, and pollutants on aquatic organisms. The current model uses GC50, the concentration of a substance that causes 50% inhibition of growth over 48 hours, as an indicator of toxicity and was trained on 1898 compounds using ordinary least square regression method.

Acute toxicity to algae: Algae, a diverse group of aquatic organisms, are commonly employed to evaluate the toxicity of chemicals. Each species of algae has its own unique advantages and disadvantages in toxicity testing. However, data on algal toxicity is relatively scarce and variable. The current model, which is based on data collected from various publications, uses EC50, which represents the concentration of a substance that causes a specific effect in 50% of the test organisms over 72 hours, as an indicator of toxicity. This data includes toxicity towards various species of algae such as *Pseudokirchneriella subcapitata*, *Desmodesmus subspicatus*, *Navicula pelliculosa*, *Synechococcus leopoliensis* and others. The model was developed by training it on 1377 compounds using the ordinary least square regression method.

Bio concentration factor: Bioaccumulation factor (BAF) or Bio concentration factor (BCF) is a measure of the accumulation of a chemical in an organism's tissue, represented as the ratio of the chemical's concentration in an organism's tissue to the chemical's concentration in the surrounding environment (usually water) at steady state. BAF or BCF values are commonly used to evaluate the potential for chemicals to accumulate in aquatic organisms and to identify persistent chemicals. High BAF or BCF values indicate a high risk of bioaccumulation and potential toxic effects. The current model, which is based on data collected from various publications, uses Log BCF measured in fish and was developed by training it on 563 compounds using ordinary least square regression method.

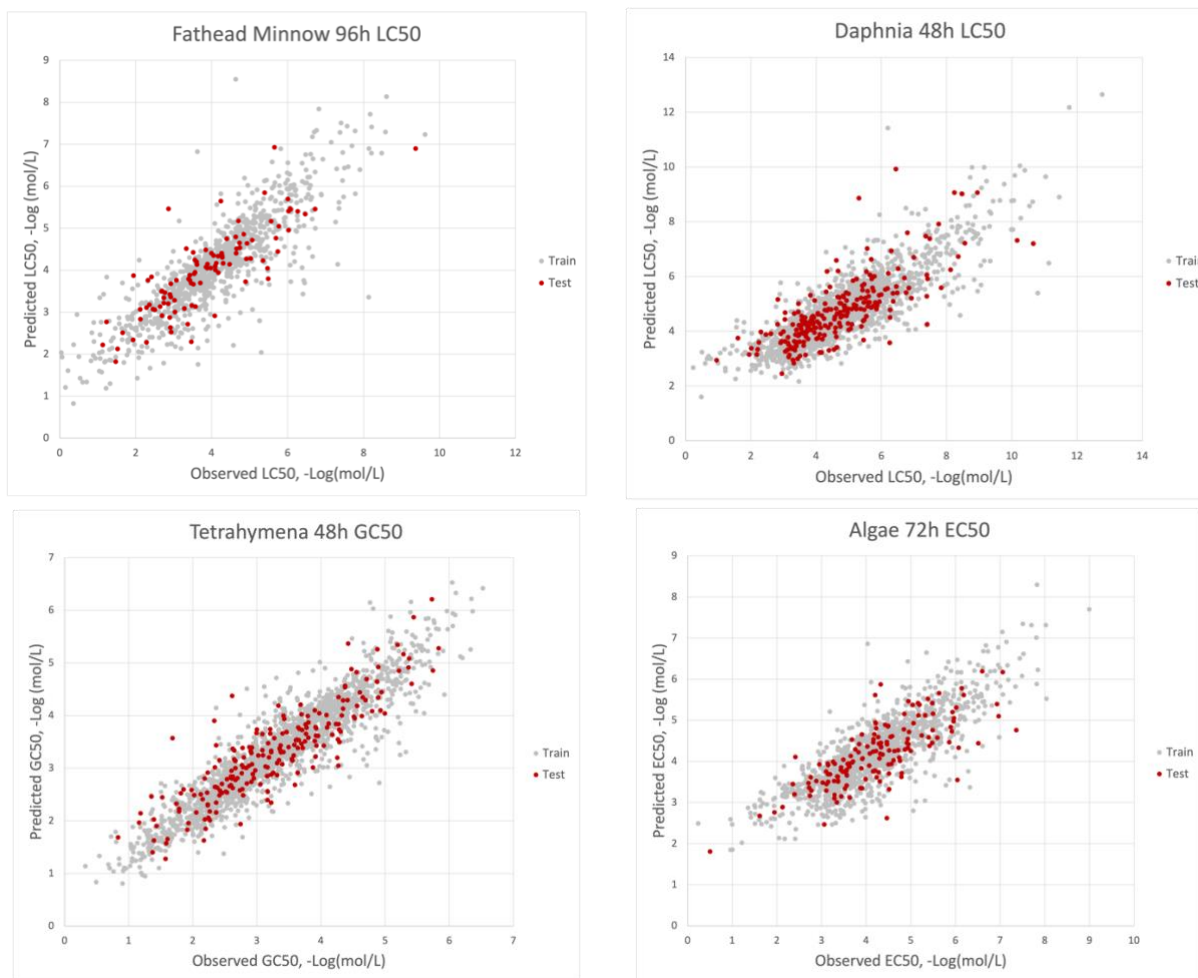
Ready biodegradability: Ready biodegradability is a measure of a chemical or substance's ability to be broken down by microorganisms in the environment. It is determined by exposing the chemical to microorganisms under controlled laboratory conditions and monitoring the degree of biodegradation over a specific period, typically 28 days. The present model uses OECD 301 ready-biodegradability test which measures biochemical oxygen demand (BOD) in aerobic aqueous medium. Chemicals with a BOD value greater than 60% are considered readily biodegradable, while those with a BOD lower than 60% are considered non-readily biodegradable. The current model, a binary classification model, is based on data from various publications, uses 1443 compounds and was developed using logistic regression method.

QSAR Flex Model Information

Soil adsorption: KOC (Organic Carbon-Sorption Coefficient) is a measure of a chemical's ability to adsorb to organic matter in soil or sediment. A high KOC means the chemical has stronger adsorption to soil, leading to lower mobility and lower toxicity. This model is trained on 651 compounds with toxicity represented by Log L/Kg, using ordinary least square regression as the methodology.

Validation of the Models

The models underwent thorough validation through internal cross-validation (CV) and by predicting both the training set and external test sets withheld during the development phase. The CV process was executed in a 10-fold manner (except the BCF model which was validated 5% out 20 times), where the model was completely rebuilt in every cycle.



QSAR Flex Model Information

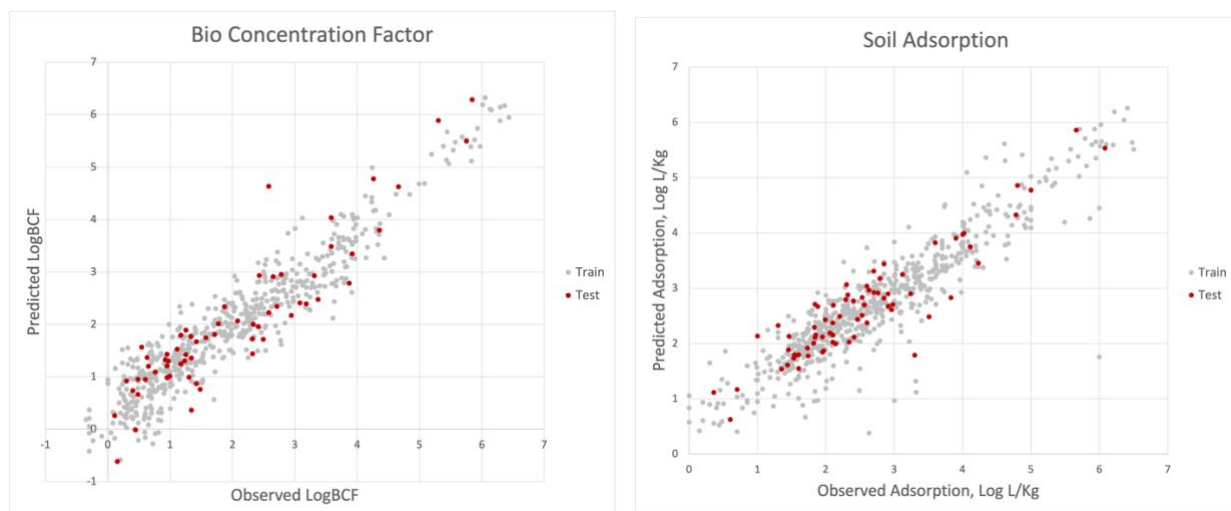


Figure 1. Comparison of experimental and predicted toxicity for the regression-based ecotoxicity models, for both the training and external test sets.

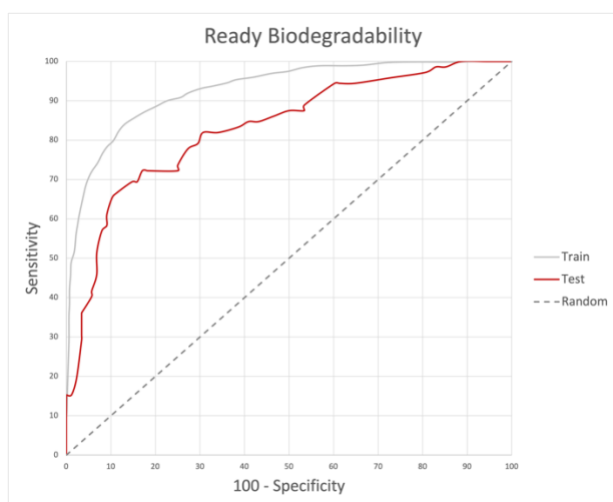


Figure 1. Performance of the ready biodegradability model in predicting results, for both the training and external test sets.

Table 2: Results of validation of the regression-based ecotoxicity models.

Model	Train		Test		CV		Train set size	Test set size
	R^2	RMSE	R^2	RMSE	R^2	RMSE		
Fathead Minnow 96h LC50	0.769	0.730	0.689	0.796	0.693	0.810	920	102
Daphnia 48h LC50	0.712	0.795	0.589	0.954	0.598	0.934	2124	235
Tetrahymena 48h GC50	0.857	0.404	0.771	0.490	0.773	0.506	1898	210
Algae 72h EC50	0.635	0.684	0.577	0.736	0.492	0.794	1377	153
Bio Concentration Factor	0.897	0.457	0.841	0.570	0.819	0.596	563	62
Soil Adsorption	0.846	0.488	0.838	0.468	0.829	0.478	651	72

R^2 – Squared correlation coefficient; $RMSE$ – Root mean squared error; CV – 10% out 10 times cross validation (except the BCF model which was validated 5% out 20 times).

Table 3: Results of validation of the classification-based model (ready biodegradability).

Model	Train			Test			CV			Train set size	Test set size
	Sens.	Spec.	AUC	Sens.	Spec.	AUC	Sens.	Spec.	AUC		
Ready Biodegradability	0.798	0.894	0.928	0.722	0.830	0.831	0.741	0.848	0.857	1443	160

Sens. – Sensitivity; Spec. – Specificity; AUC – Area under the Receiver Operating Characteristic (ROC) curve; CV – 10% out 10 times cross validation.

Determination of Domain Applicability

When a test chemical is evaluated through one of the above-mentioned QSAR models, the initial step is to determine if the chemical already exists in the model's database of training compounds. If it does, the corresponding experimental activity/property will be provided. If not, the chemical is then assessed for domain applicability, where it is compared against a database of small molecular fragments (derived from the training set). If a certain number of fragments are absent in the chemical, it is deemed to be out-of-domain.

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